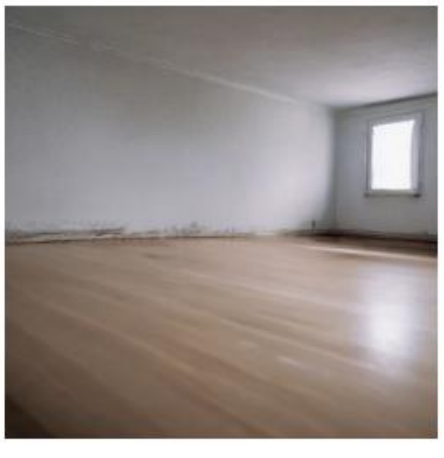
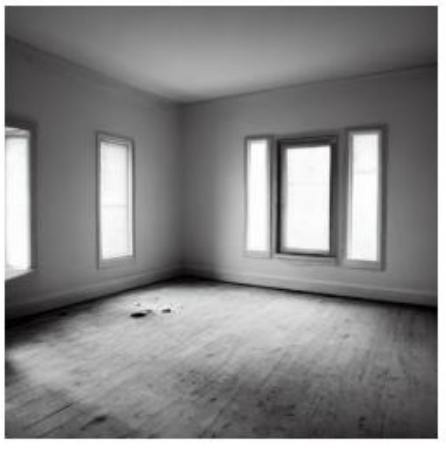
Baseline: DDIM 50 steps, w = 7.5, 512x512 (base) - prompts 1 to 5

	COCO photo (reference)	seed = gen_seed + 0	seed = gen_seed + 100	seed = gen_seed + 200
prompt_01: there is a large book shelf in this living room				
prompt_02: A man riding on the back of a brown horse through a lush green field.				
prompt_03: A hot dog with sour kraut on a plate.				
prompt_04: there is various healthy vegetables on a plate				
prompt_05: A house is empty with no one in it.				

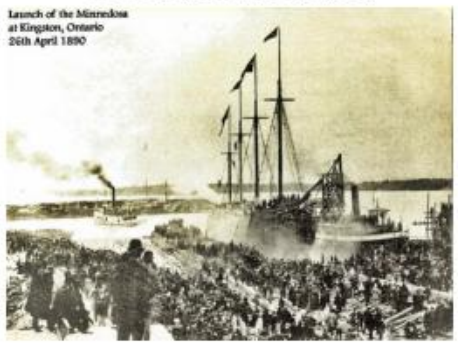
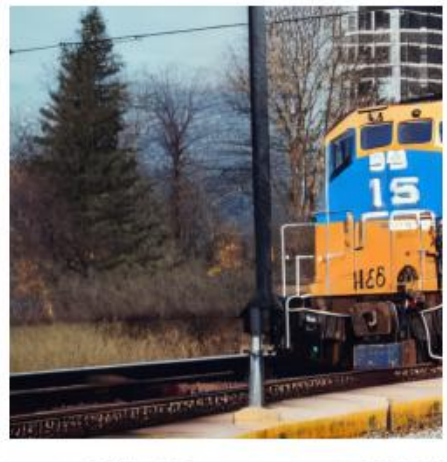
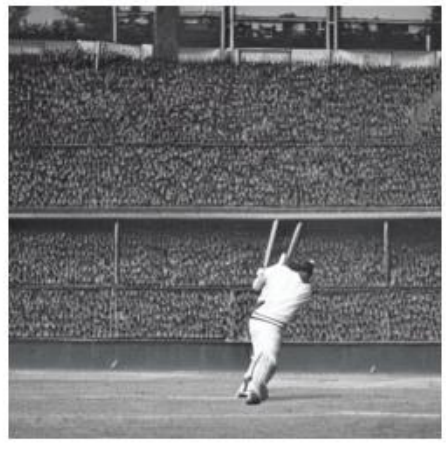
Baseline: DDIM 50 steps, w = 7.5, 512x512 (base) - prompts 6 to 10

	COCO photo (reference)	seed = gen_seed + 0	seed = gen_seed + 100	seed = gen_seed + 200
prompt_06: A gray cat laying on top of a wooden book shelf.				
prompt_07: A MAN IS PLAYING ON THE BEACH WITH A FRIEND				
prompt_08: A green truck cake sitting on top of a table.				
prompt_09: A red double decker bus parked in front of a bus depot.				
prompt_10: Three people sitting down together for a meal.				

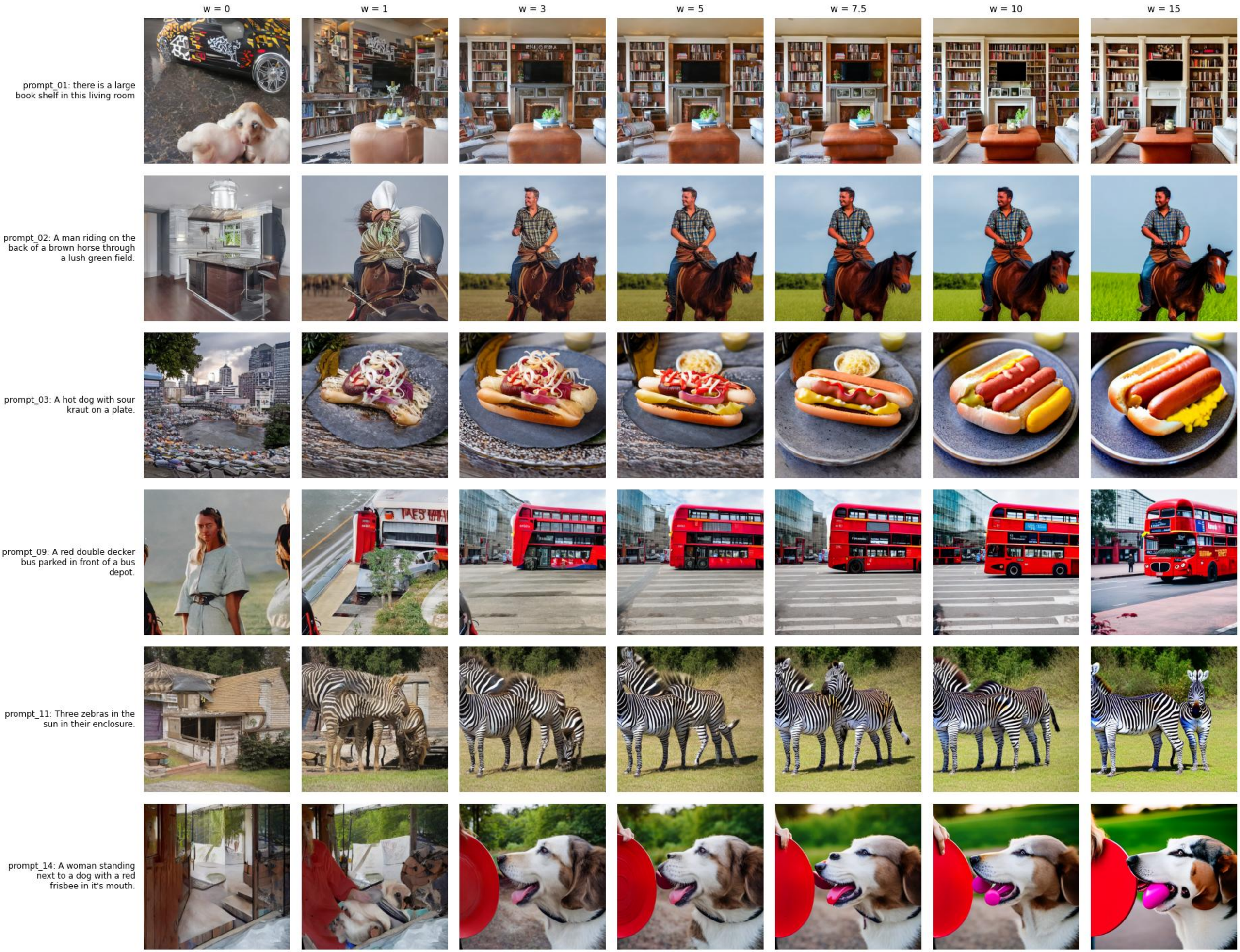
Baseline: DDIM 50 steps, w = 7.5, 512x512 (base) - prompts 11 to 15

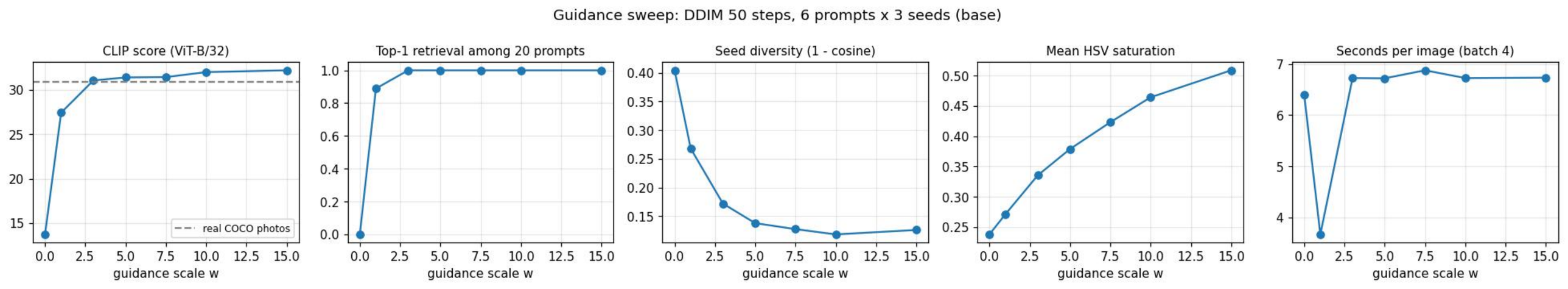
	COCO photo (reference)	seed = gen_seed + 0	seed = gen_seed + 100	seed = gen_seed + 200
prompt_11: Three zebras in the sun in their enclosure.				
prompt_12: a doll is sitting next to some pastries on a tray				
prompt_13: A pizza buffet with one slice taken out of a pizza.				
prompt_14: A woman standing next to a dog with a red frisbee in it's mouth.				
prompt_15: A traffic light is flashing the color green.				

Baseline: DDIM 50 steps, w = 7.5, 512x512 (base) - prompts 16 to 20

	COCO photo (reference)	seed = gen_seed + 0	seed = gen_seed + 100	seed = gen_seed + 200
prompt_16: a picture of an old boat with a lot of people around it				
prompt_17: A man fixing the inside of a toilet				
prompt_18: A group of men in blue and red jerseys eating from a cake.				
prompt_19: A man with a brown bag and blue jacket stands by a train that is moving.				
prompt_20: A player swings a cricket bat on the playing field.				

Guidance scale: DDIM 50 steps, seed = gen_seed + 0 (base)




Guidance sweep: DDIM 50 steps, 6 prompts x 3 seeds (base)
CLIP score (ViT-B/32)
real COCO photos
guidance scale w
Top-1 retrieval among 20 prompts
guidance scale w
Seed diversity (1 - cosine)
guidance scale w
Mean HSV saturation
guidance scale w
Seconds per image (batch 4)
guidance scale w

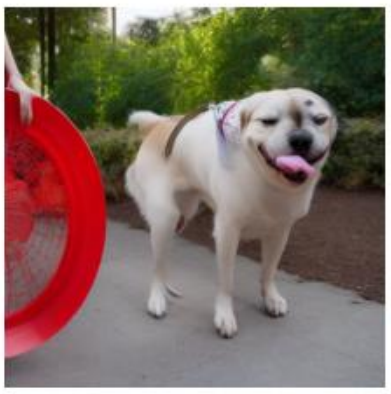
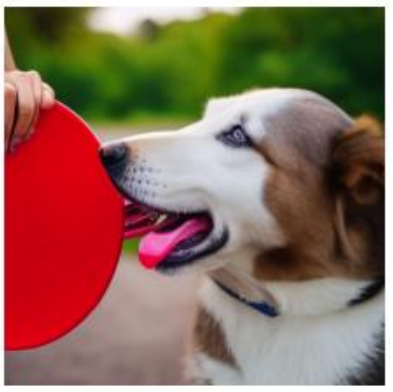
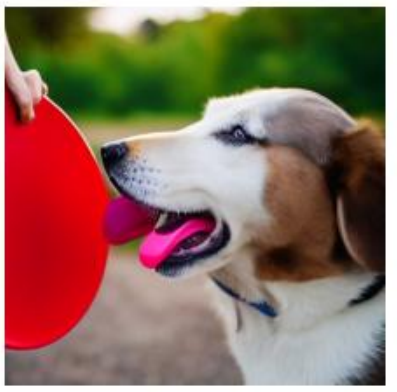
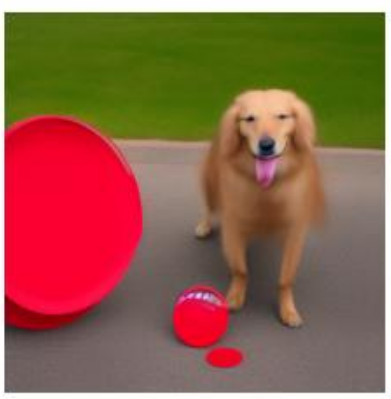
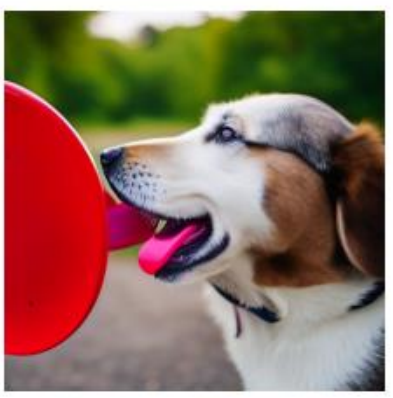
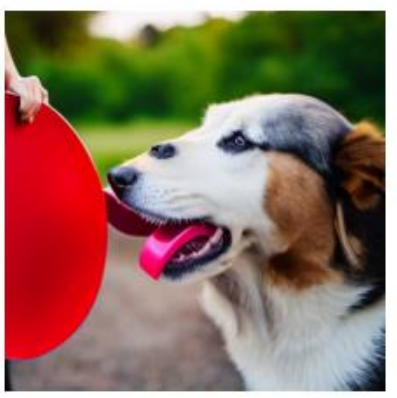
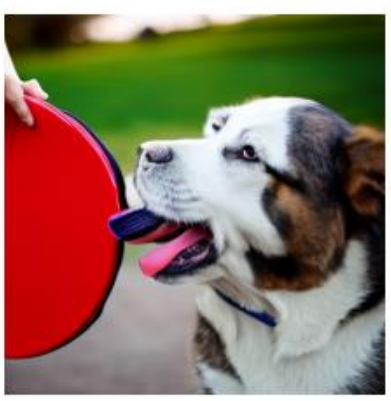
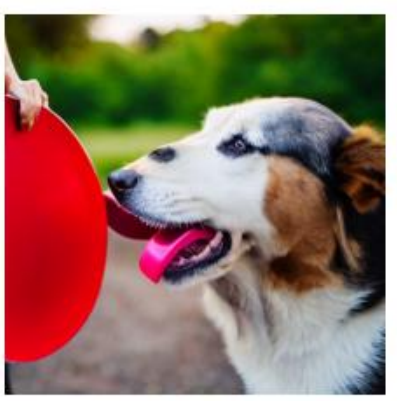
Seeds at w = 1, 7.5 and 15: DDIM 50 steps (base)

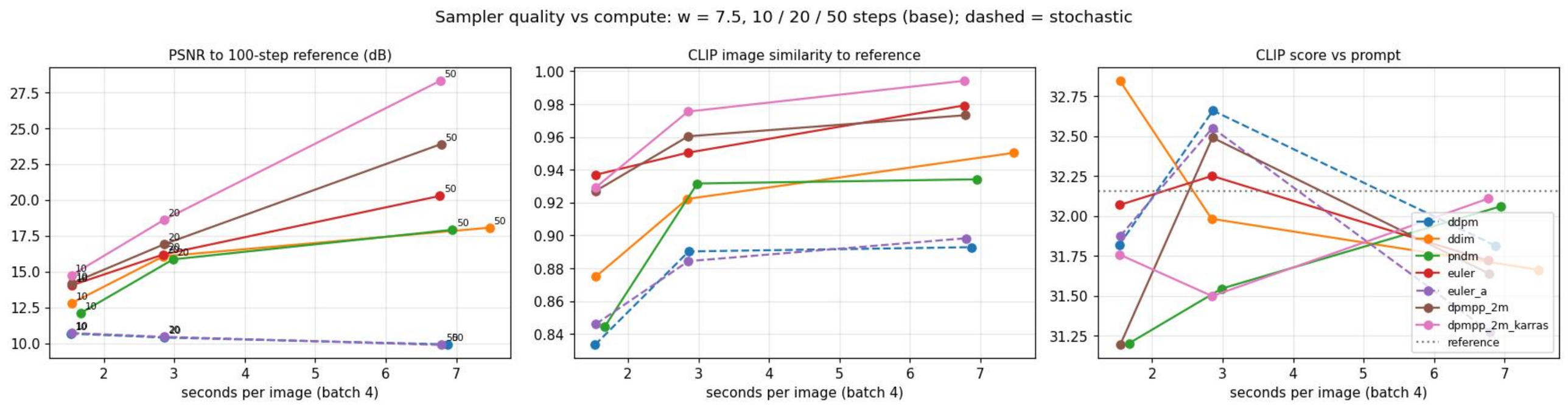
	seed = gen_seed + 0	seed = gen_seed + 100	seed = gen_seed + 200
w = 1 | prompt_02: A man riding on the back of a brown horse through a lush green field.			
w = 7.5 | prompt_02: A man riding on the back of a brown horse through a lush green field.			
w = 15 | prompt_02: A man riding on the back of a brown horse through a lush green field.			
w = 1 | prompt_09: A red double decker bus parked in front of a bus depot.			
w = 7.5 | prompt_09: A red double decker bus parked in front of a bus depot.			
w = 15 | prompt_09: A red double decker bus parked in front of a bus depot.			

prompt_02: A man riding on the back of a brown horse through a lush green field. (w = 7.5, base)

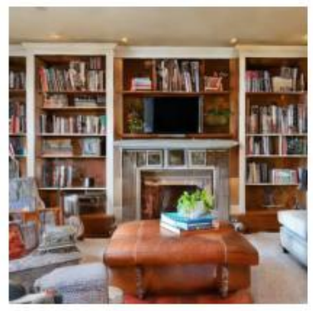
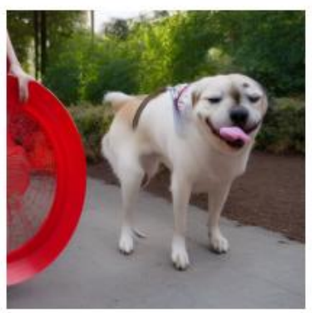
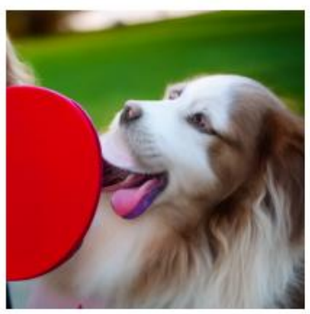
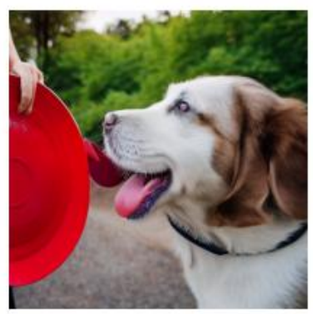
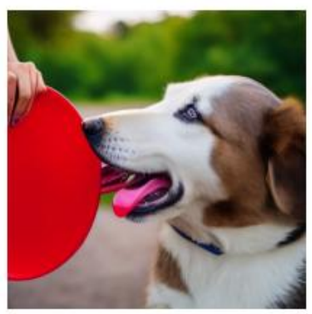
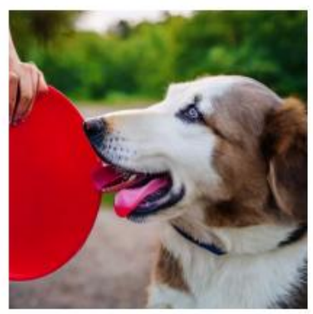
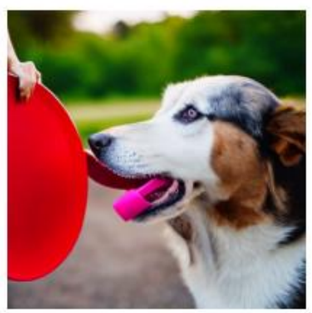
	10 steps	20 steps	50 steps	reference: DPM++ 2M, 100
ddpm				
ddim				
pndm				
euler				
euler_a				
dpmpp_2m				
dpmpp_2m_karras				

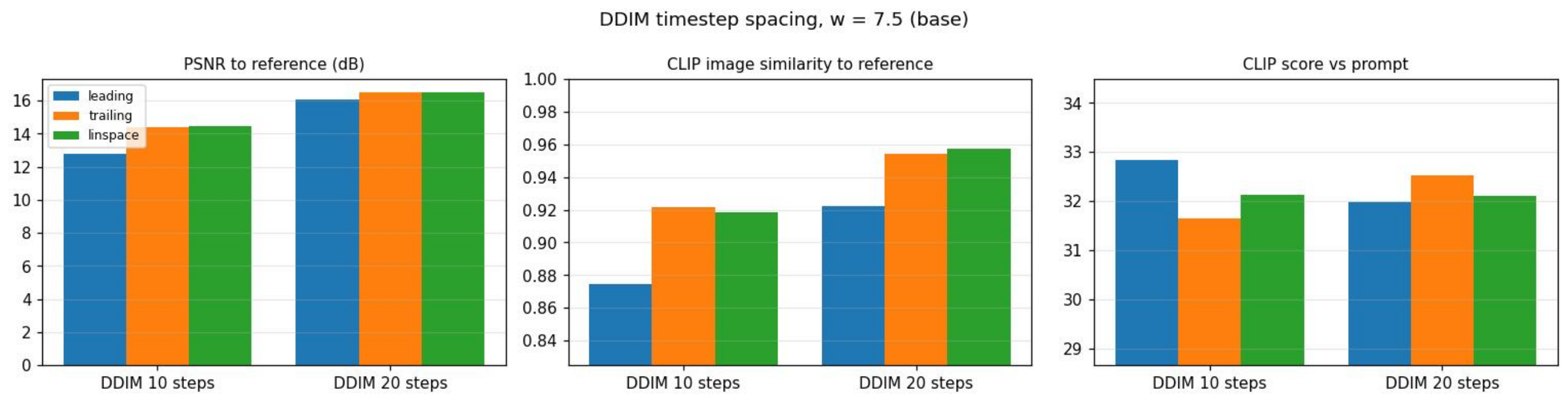
prompt_14: A woman standing next to a dog with a red frisbee in it's mouth. (w = 7.5, base)

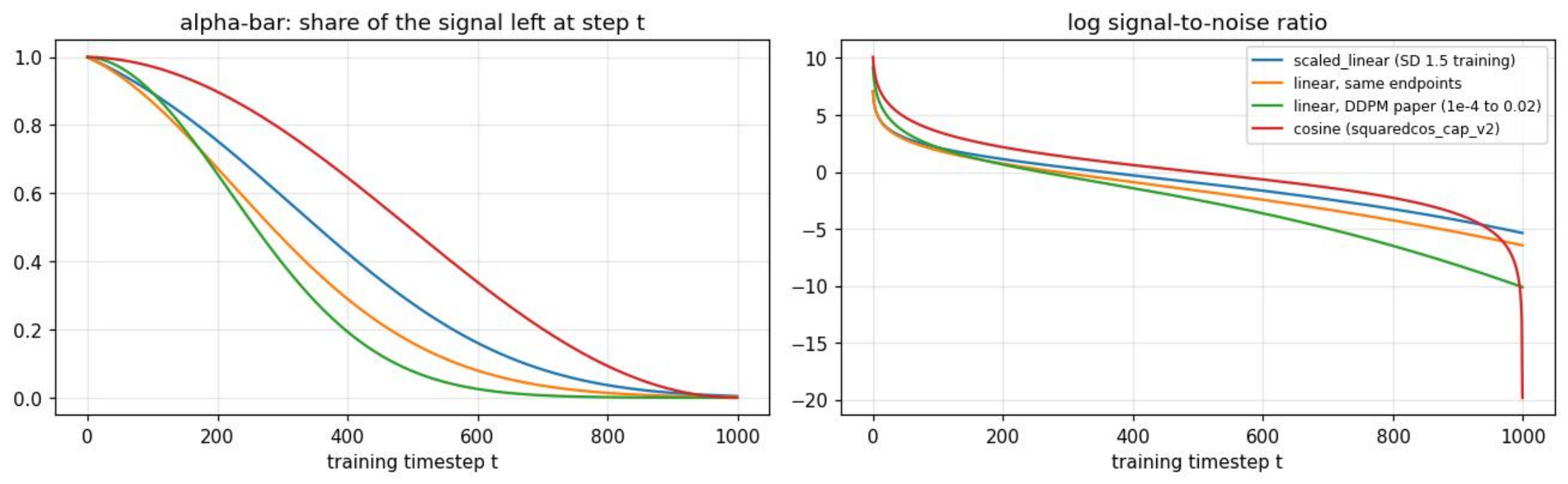
	10 steps	20 steps	50 steps	reference: DPM++ 2M, 100
ddpm				
ddim				
pndm				
euler				
euler_a				
dpmpp_2m				
dpmpp_2m_karras				


Sampler quality vs compute: w = 7.5, 10 / 20 / 50 steps (base); dashed = stochastic
PSNR to 100-step reference (dB)
CLIP image similarity to reference
CLIP score vs prompt
seconds per image (batch 4)
seconds per image (batch 4)
seconds per image (batch 4)
ddpm
ddim
pndm
euler
euler_a
dpmpp_2m
dpmpp_2m_karras
reference

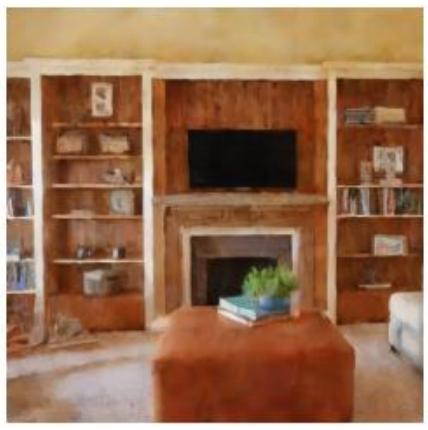
DDIM timestep spacing, w = 7.5 (base)

	leading, 10 steps	trailing, 10 steps	linspace, 10 steps	leading, 20 steps	trailing, 20 steps	linspace, 20 steps	reference: DPM++ 2M, 100
prompt_01: there is a large book shelf in this living room							
prompt_02: A man riding on the back of a brown horse through a lush green field.							
prompt_03: A hot dog with sour kraut on a plate.							
prompt_09: A red double decker bus parked in front of a bus depot.							
prompt_11: Three zebras in the sun in their enclosure.							
prompt_14: A woman standing next to a dog with a red frisbee in it's mouth.							


DDIM timestep spacing, w = 7.5 (base)
PSNR to reference (dB)
leading
trailing
linspace
0
2
4
6
8
10
12
14
16
DDIM 10 steps
DDIM 20 steps
CLIP image similarity to reference
0.84
0.86
0.88
0.90
0.92
0.94
0.96
0.98
1.00
DDIM 10 steps
DDIM 20 steps
CLIP score vs prompt
29
30
31
32
33
34
DDIM 10 steps
DDIM 20 steps


alpha-bar: share of the signal left at step t
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
training timestep t
log signal-to-noise ratio
10
5
0
−5
−10
−15
−20
scaled_linear (SD 1.5 training)
linear, same endpoints
linear, DDPM paper (1e-4 to 0.02)
cosine (squaredcos_cap_v2)
0
200
400
600
800
1000
training timestep t

Sampling with a beta schedule the model was not trained on: DDIM 50, w = 7.5 (base)

	scaled_linear (trained)	beta-linear	beta-linear-ddpm	beta-cosine
prompt_01: there is a large book shelf in this living room				
prompt_02: A man riding on the back of a brown horse through a lush green field.				
prompt_03: A hot dog with sour kraut on a plate.				
prompt_09: A red double decker bus parked in front of a bus depot.				
prompt_11: Three zebras in the sun in their enclosure.				
prompt_14: A woman standing next to a dog with a red frisbee in it's mouth.				